Experiment 5 - Functions

Python Lab Manual — Program and Output

EX. NO: 5(a) — Factorial using a function

PROGRAM

```
def factorial(n):
    if n == 0:
        return 1
    else:
        return n * factorial(n - 1)

n = int(input("Input a number to compute the factorial: "))
print(factorial(n))
```

OUTPUT

```
Input a number to compute the factorial: 4
24
```

EX. NO: 5(b) — Find the largest number in a list

PROGRAM

```
def max_num_in_list(values):
    largest = values[0]
    for a in values:
        if a > largest:
            largest = a
    return largest

print(max_num_in_list([1, 2, -8, 0]))
```

OUTPUT

```
2
```

EX. NO: 5(c) — Area of shapes

PROGRAM

```
def calculate_area(name):
    name = name.lower()

    if name == "rectangle":
        l = int(input("Enter rectangle's length: "))
        b = int(input("Enter rectangle's breadth: "))
        print(f"The area of rectangle is {l * b}.")
    elif name == "square":
        s = int(input("Enter square's side length: "))
        print(f"The area of square is {s * s}.")
    elif name == "triangle":
        h = int(input("Enter triangle's height length: "))
        b = int(input("Enter triangle's breadth length: "))
        print(f"The area of triangle is {0.5 * b * h}.")
    elif name == "circle":
        r = int(input("Enter circle's radius length: "))
        pi = 3.14
        print(f"The area of circle is {pi * r * r}.")
    elif name == "parallelogram":
        b = int(input("Enter parallelogram's base length: "))
        h = int(input("Enter parallelogram's height length: "))
        print(f"The area of parallelogram is {b * h}.")
    else:
        print("Sorry! This shape is not available")

print("Calculate Shape Area")
name = input("Enter the name of shape whose area you want to find: ")
calculate_area(name)
```

OUTPUT

```
Calculate Shape Area
Enter the name of shape whose area you want to find: rectangle
Enter rectangle's length: 10
Enter rectangle's breadth: 15
The area of rectangle is 150.
```